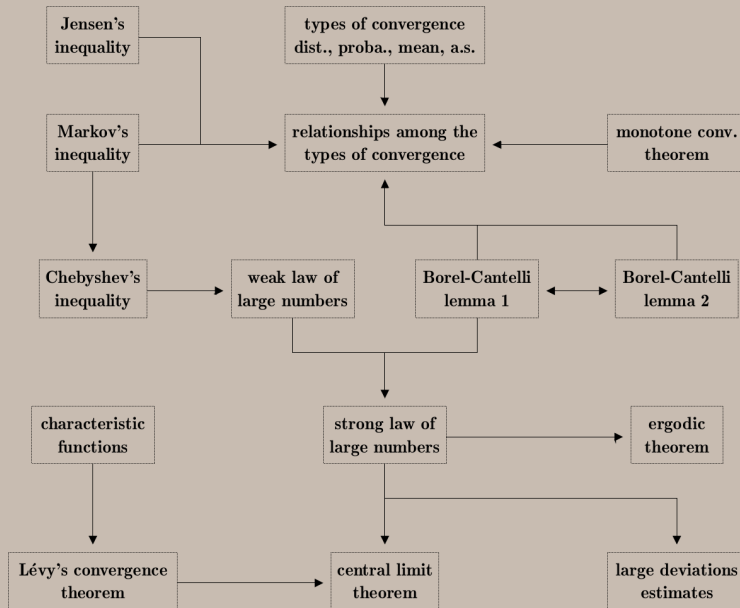
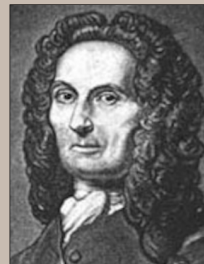
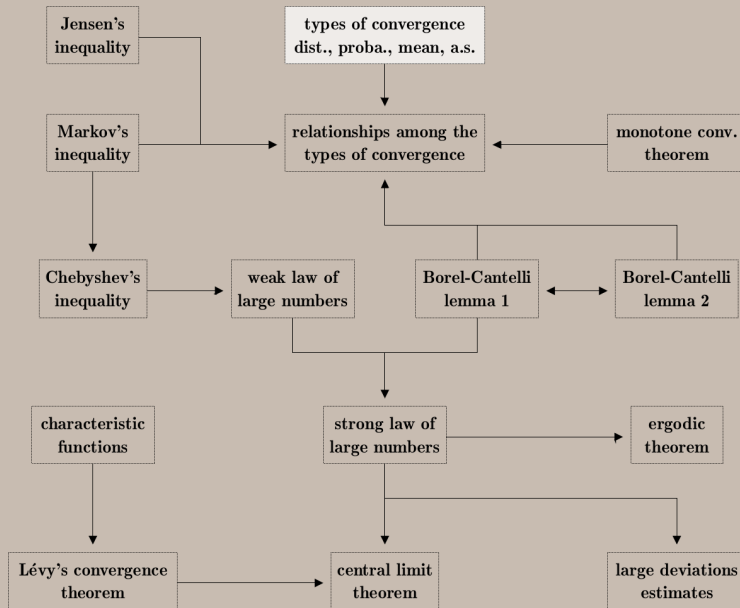




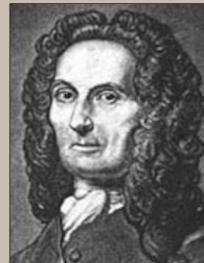
Jakob Bernoulli



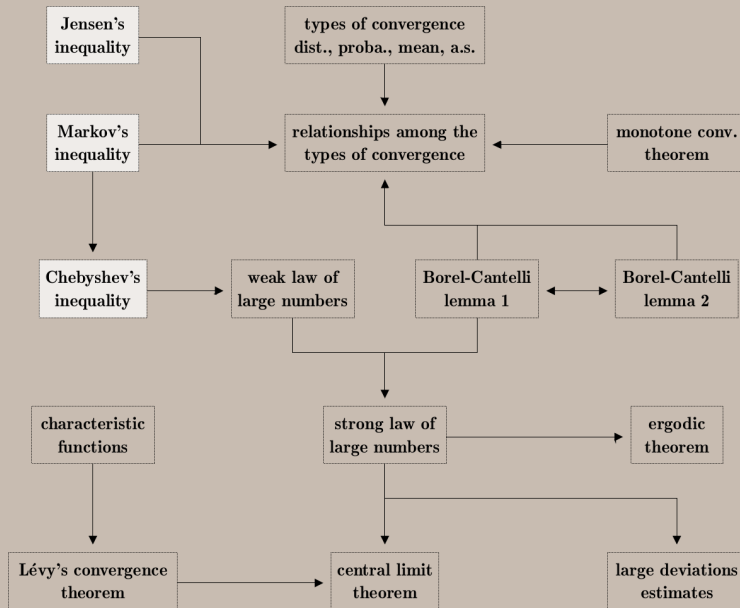
Abraham de Moivre



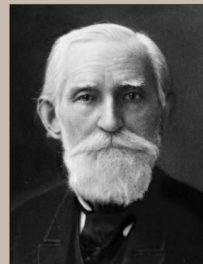
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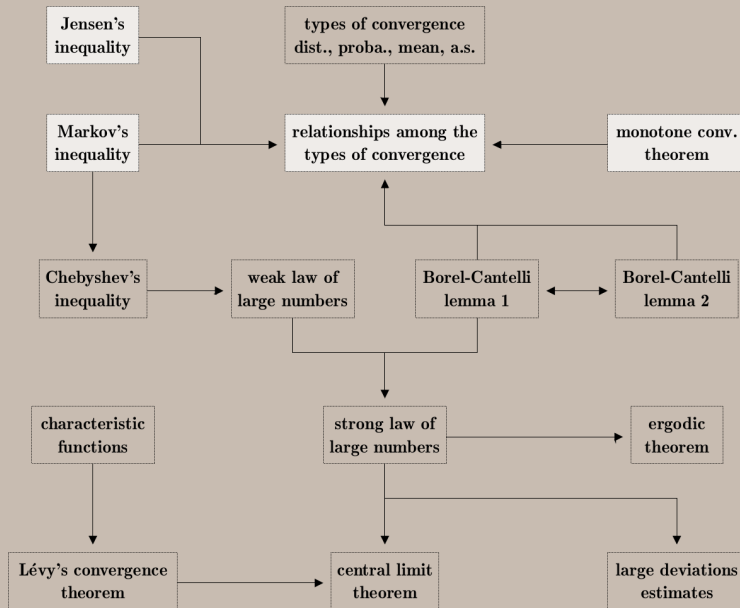
Abraham de Moivre



Andrey Markov



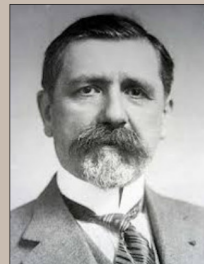
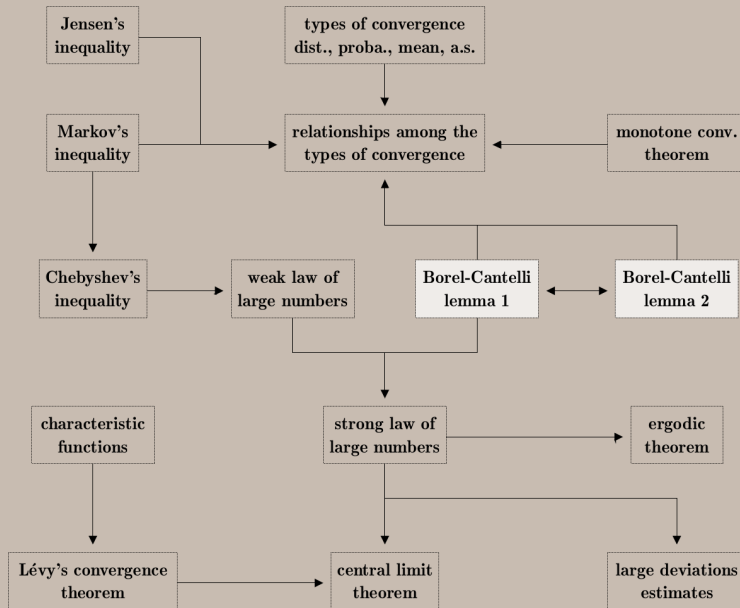
Pafnuty Chebyshev



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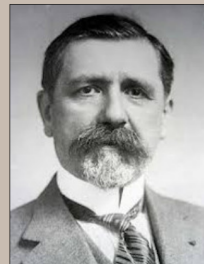
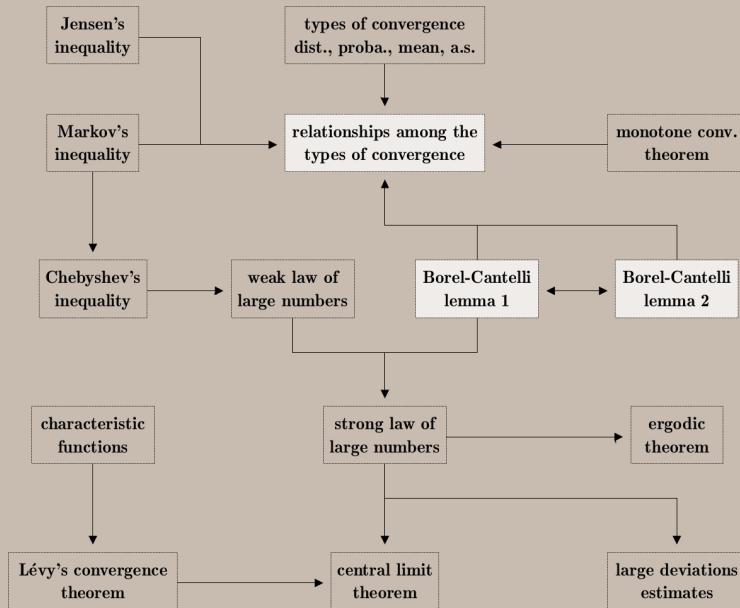
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Emile Borel



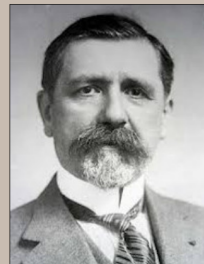
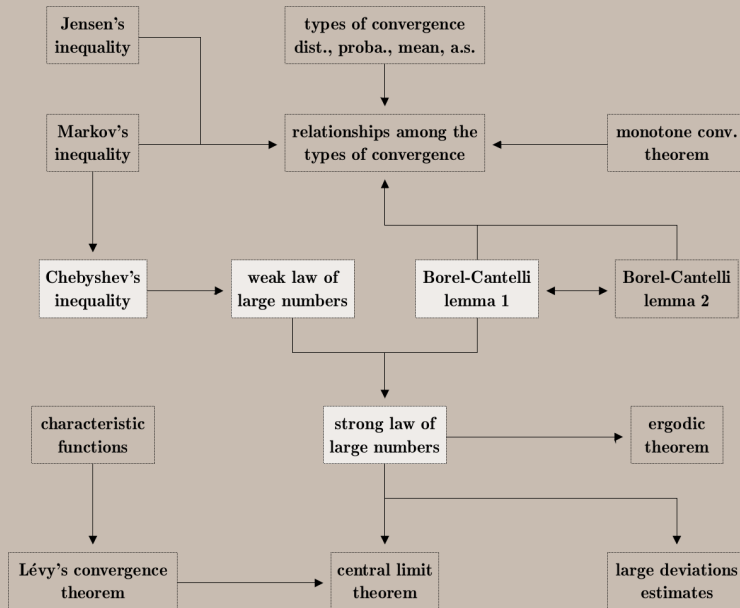
Francesco Cantelli



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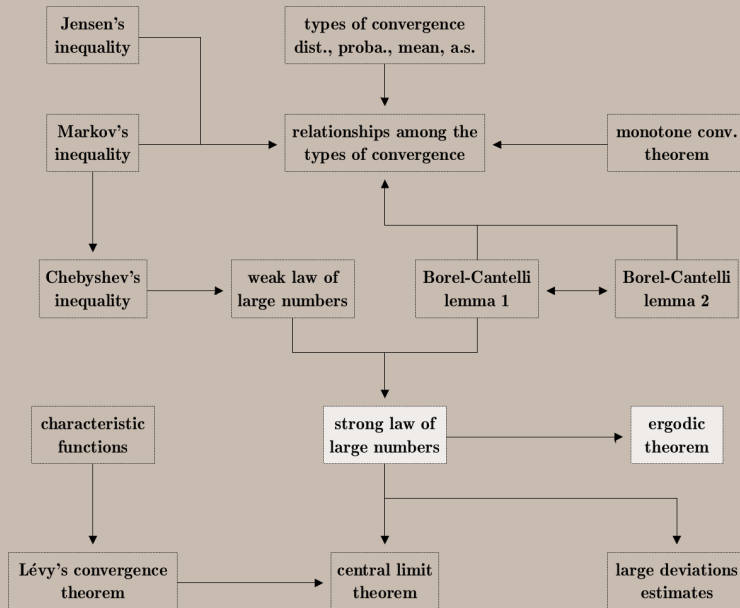
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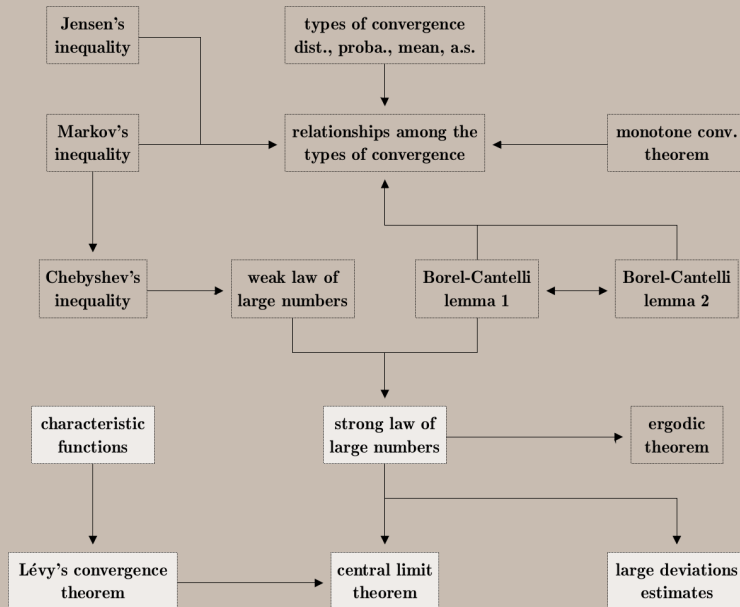
Francesco Cantelli



George Birkhoff



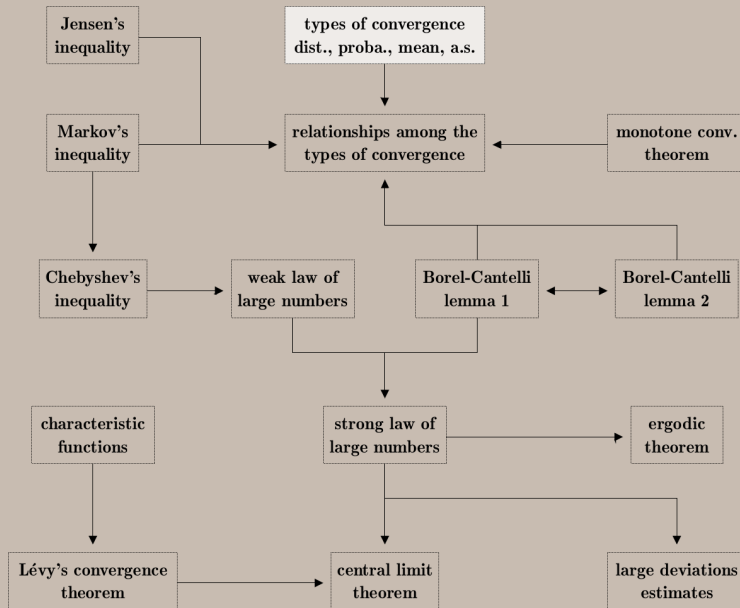
John von Neumann



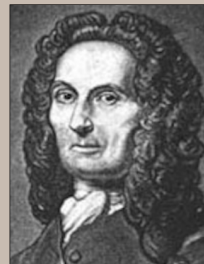
Paul Lévy



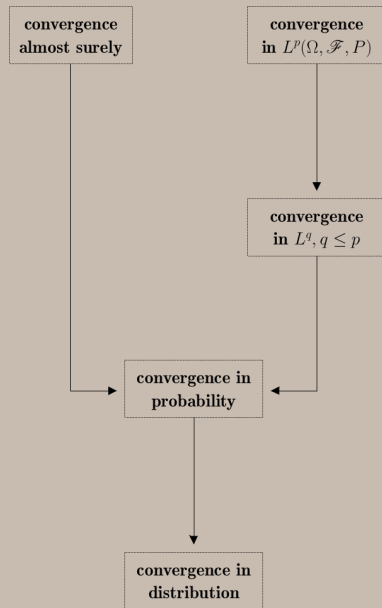
Srinivasa Varadhan

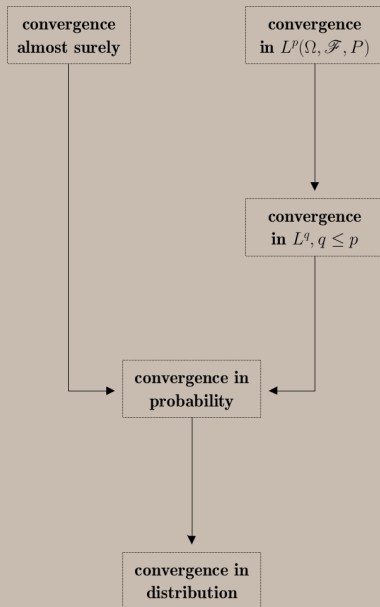


Jakob Bernoulli

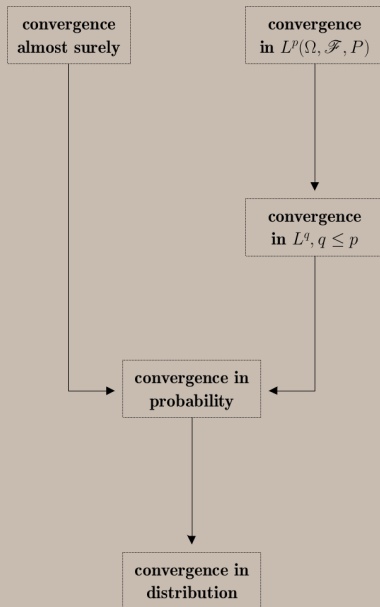


Abraham de Moivre





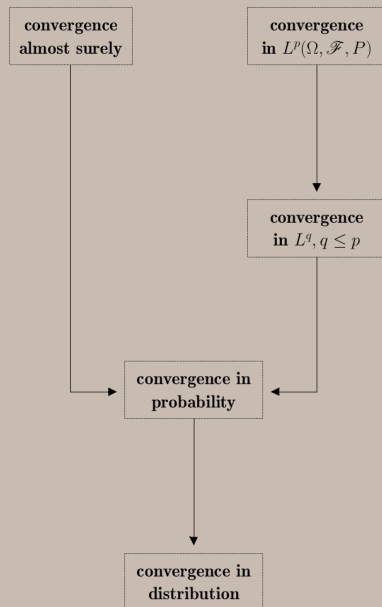
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$$P(X_n \rightarrow X) = P(\{\omega \in \Omega : X_n(\omega) \rightarrow X(\omega)\}) = 1.$$



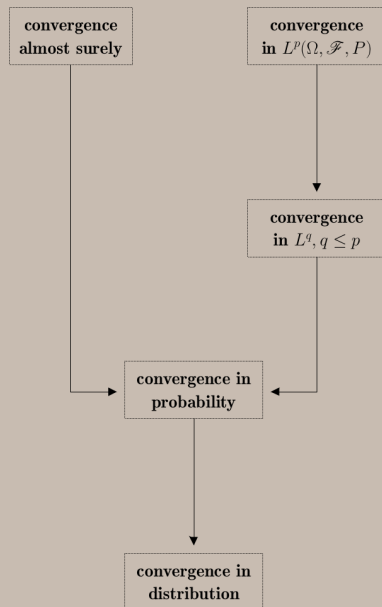
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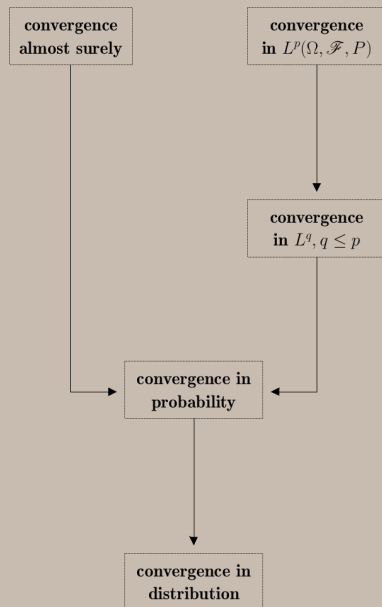
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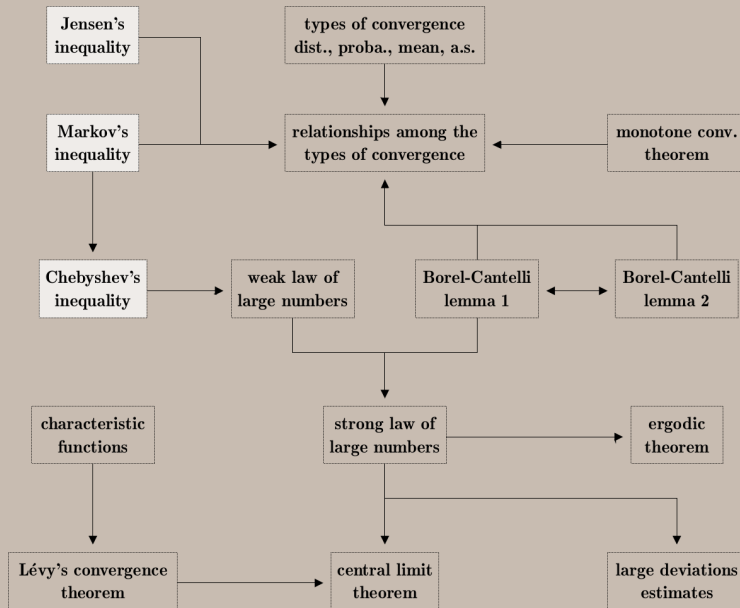
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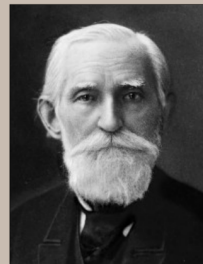
- 4) Convergence in **distribution** ($X_n \xrightarrow{d} X$)

$$F_{X_n}(x) = P(X_n \leq x) \rightarrow F_X(x) = P(X \leq x)$$

on the continuity set of F_X .



Andrey Markov

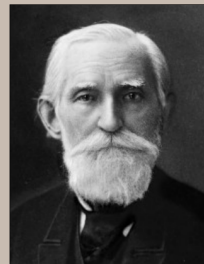


Pafnuty Chebyshev

Markov's inequality. $P(X \geq \epsilon) \leq E(X)/\epsilon$ for all $X \geq 0$ and $\epsilon > 0$.



Andrey Markov



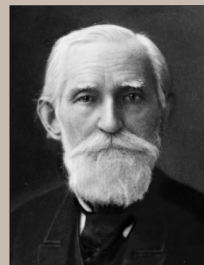
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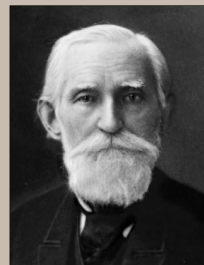
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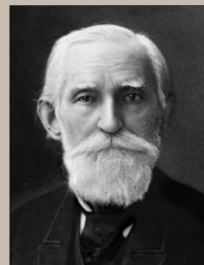
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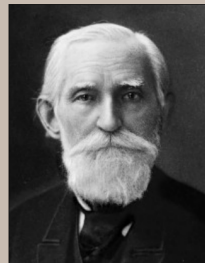
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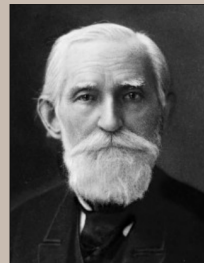
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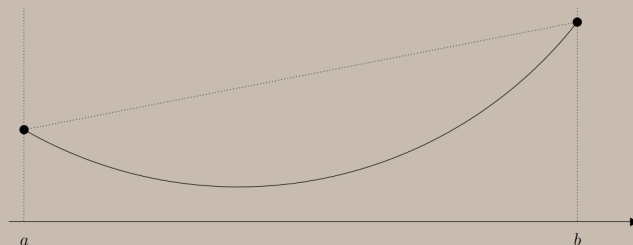
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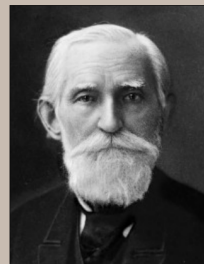
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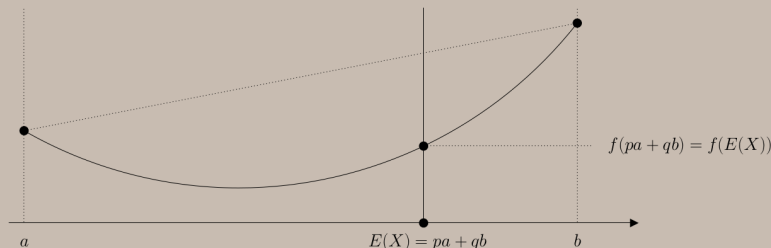
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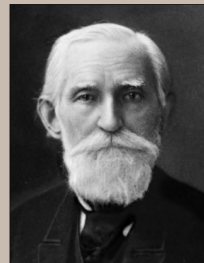
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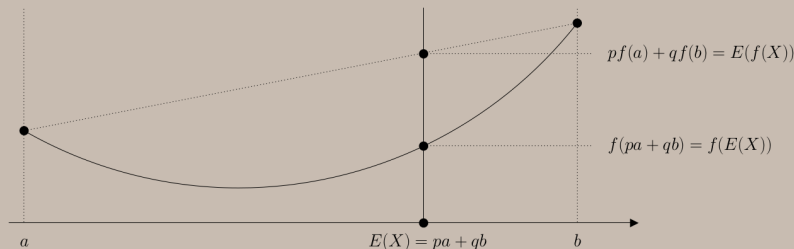
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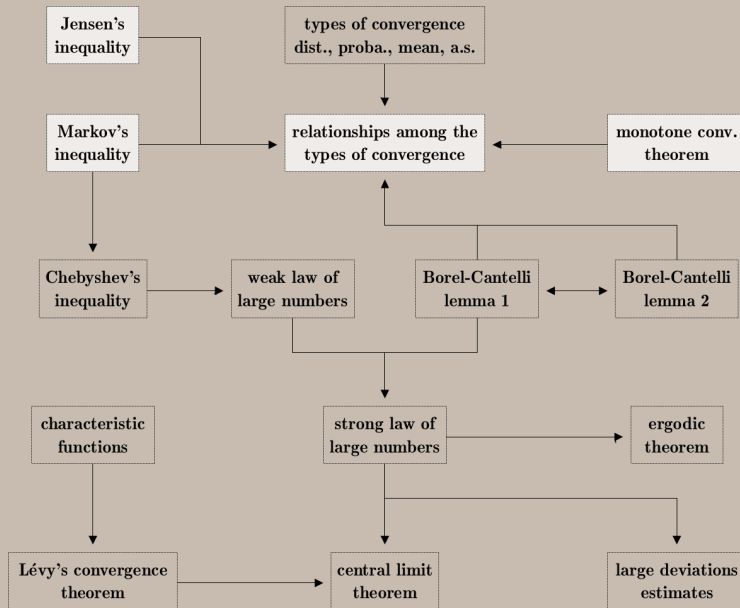


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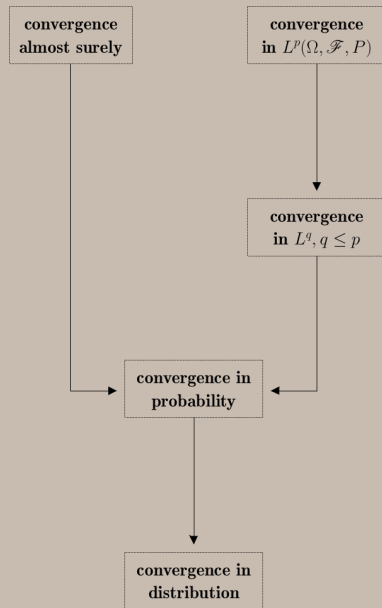


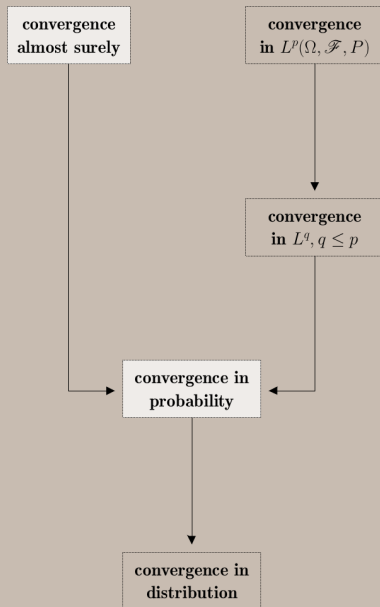


Andrey Markov



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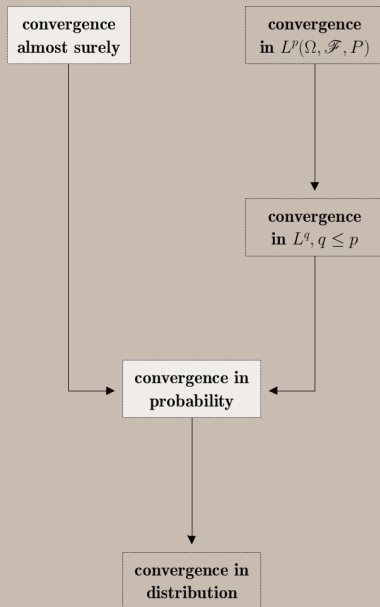




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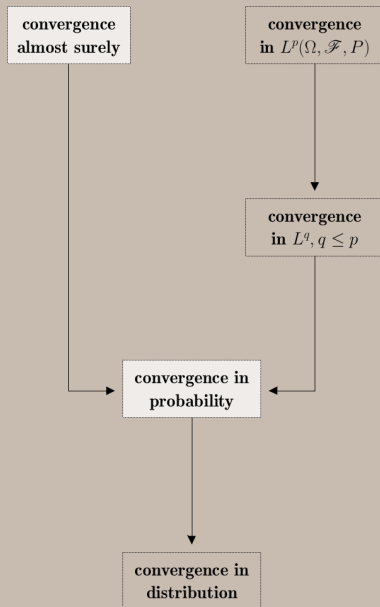


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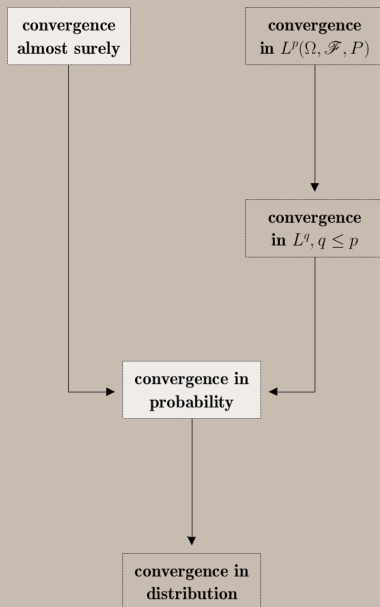
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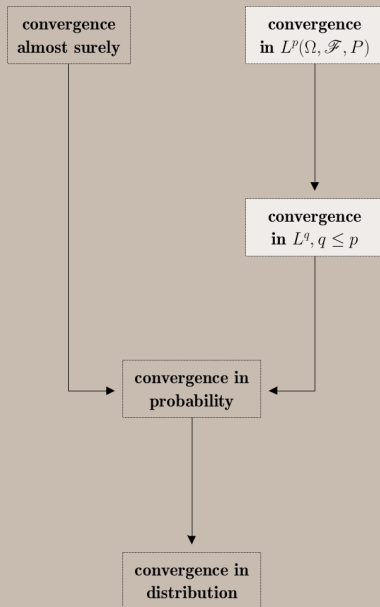
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Monotone convergence theorem $\implies$

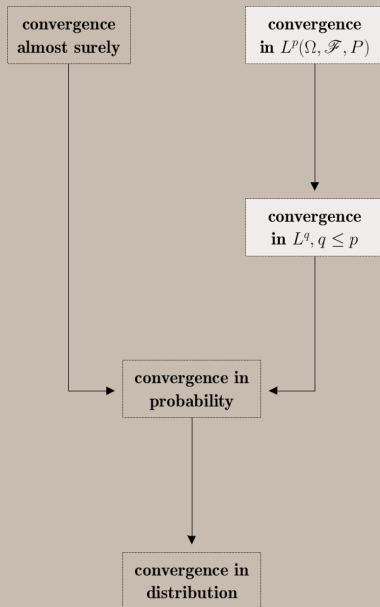
$$\begin{aligned} \lim_{n \rightarrow \infty} P(|X_n - X| \geq \epsilon) &= \lim_{n \rightarrow \infty} P(A_n) \\ &\leq \lim_{n \rightarrow \infty} P(\bigcup_{k \geq n} A_k) \\ &\stackrel{\text{MCT}}{=} P(\lim_{n \rightarrow \infty} \bigcup_{k \geq n} A_k) \\ &= P(\limsup_{n \rightarrow \infty} A_n) = 0. \end{aligned}$$



Definition. $X_n \xrightarrow{L^p} X \iff (E|X_n - X|^p)^{1/p} \rightarrow 0$.

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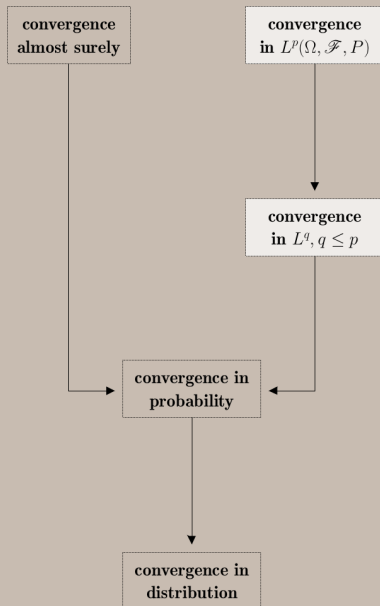


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Proof. **Jensen's ineq.** with $f(x) = x^{p/q}$ convex



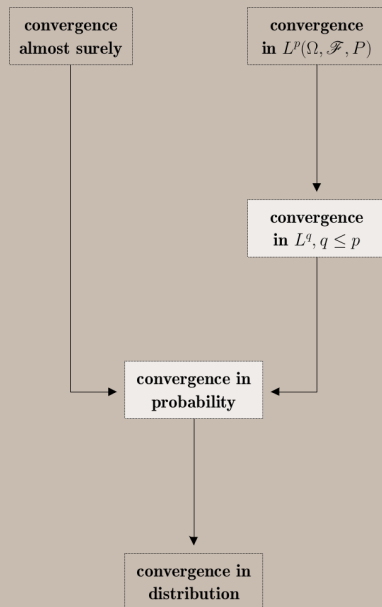
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Proof. **Jensen's ineq.** with $f(x) = x^{p/q}$ convex $\implies$

$$\begin{aligned}
 \|X_n - X\|_q &= (E|X_n - X|^q)^{1/q} \\
 &= (f(E|X_n - X|^q))^{1/p} \\
 &\stackrel{\text{JI}}{\leq} (E(f(|X_n - X|^q)))^{1/p} \\
 &= (E|X_n - X|^p)^{1/p} = \|X_n - X\|_p \rightarrow 0.
 \end{aligned}$$



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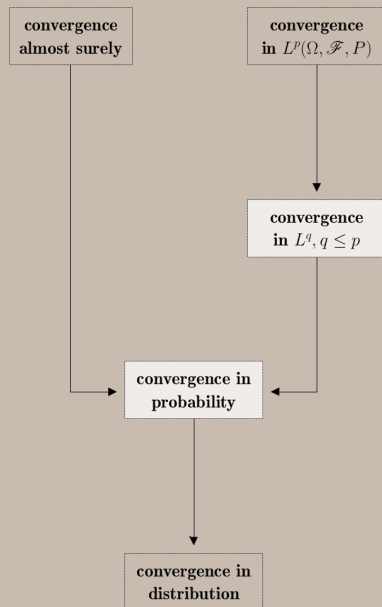
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Theorem. $X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{L^q} X$ for all $p > q \geq 1$.

Proof. **Jensen's ineq.** with $f(x) = x^{p/q}$ convex $\implies$

$$\begin{aligned}
 \|X_n - X\|_q &= (E|X_n - X|^q)^{1/q} \\
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 &\stackrel{\text{JI}}{\leq} (E(f(|X_n - X|^q)))^{1/p} \\
 &= (E|X_n - X|^p)^{1/p} = \|X_n - X\|_p \rightarrow 0.
 \end{aligned}$$

Theorem. $X_n \xrightarrow{L^p} X \implies X_n \xrightarrow{p} X$.



Definition. $X_n \xrightarrow{L^p} X \iff (E|X_n - X|^p)^{1/p} \rightarrow 0$.

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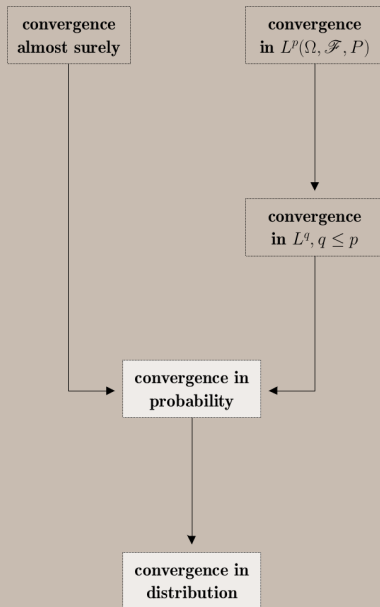
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Proof. Let $\epsilon > 0$. **Markov's inequality** $\implies$

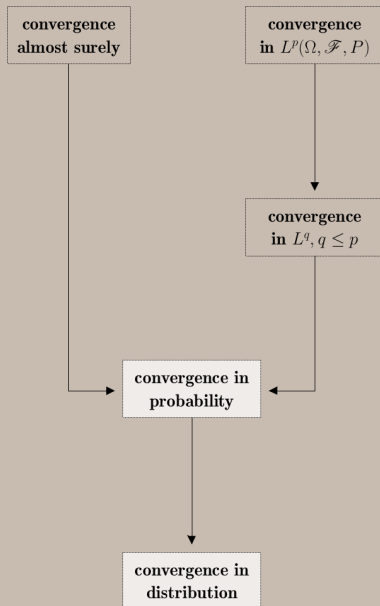
$$\begin{aligned}
 P(|X_n - X| \geq \epsilon) &\stackrel{\text{MI}}{\leq} E|X_n - X|/\epsilon \\
 &\leq \|X_n - X\|_p/\epsilon \rightarrow 0.
 \end{aligned}$$



Definition. $X_n \xrightarrow{p} X \iff P(|X_n - X| \geq \epsilon) \rightarrow 0$.

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Theorem. $X_n \xrightarrow{p} X \implies X_n \xrightarrow{d} X$.



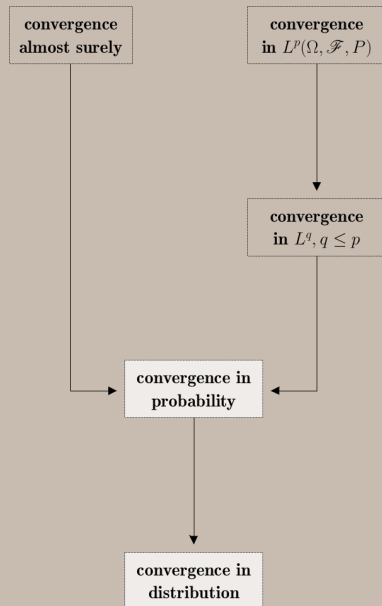
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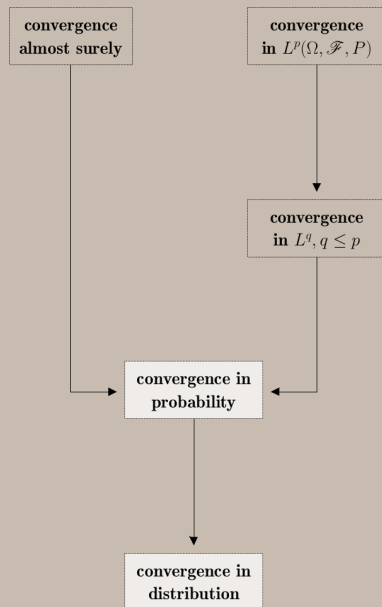
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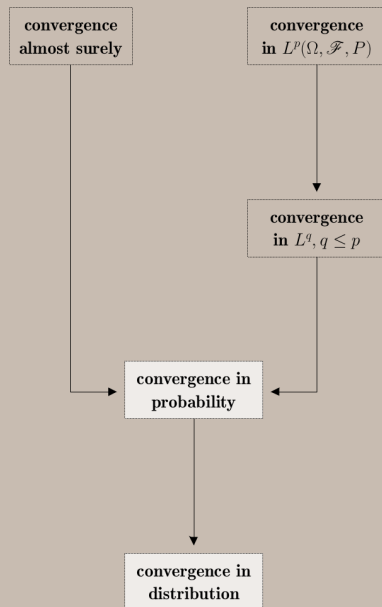
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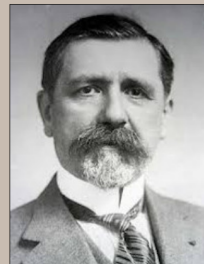
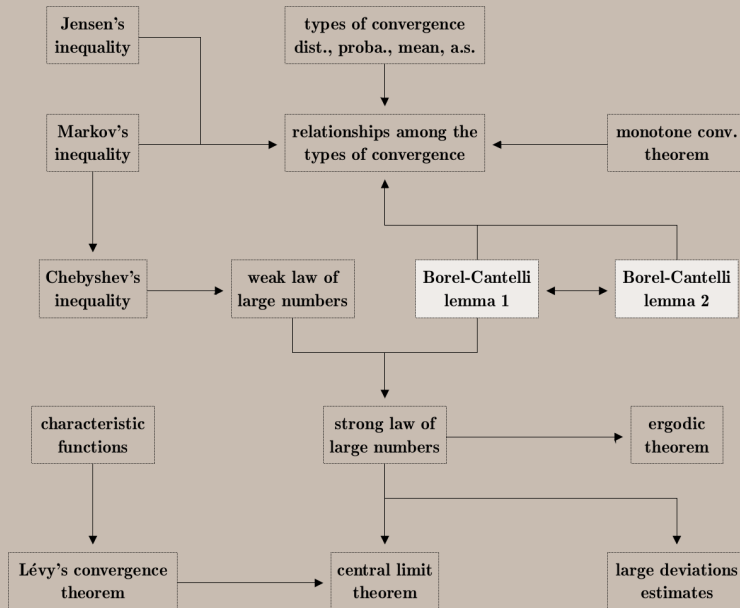
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Assume F_X is continuous at $x \implies$

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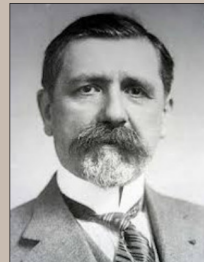
Emile Borel



Francesco Cantelli

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Emile Borel

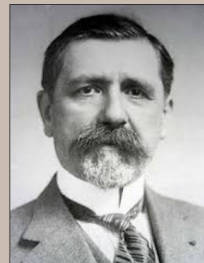


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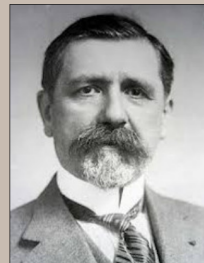
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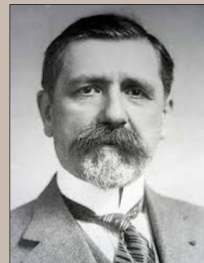
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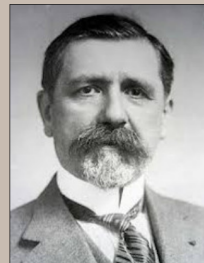
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Emile Borel



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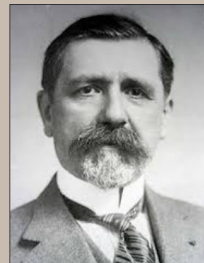
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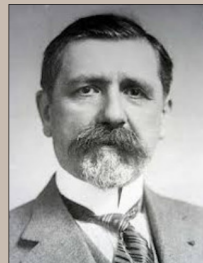
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$$2) \text{ Independence } \implies \text{ as } m \rightarrow \infty,$$

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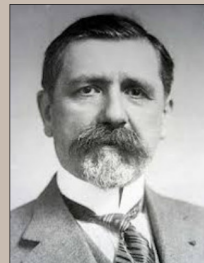
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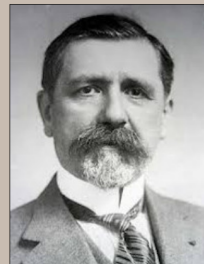
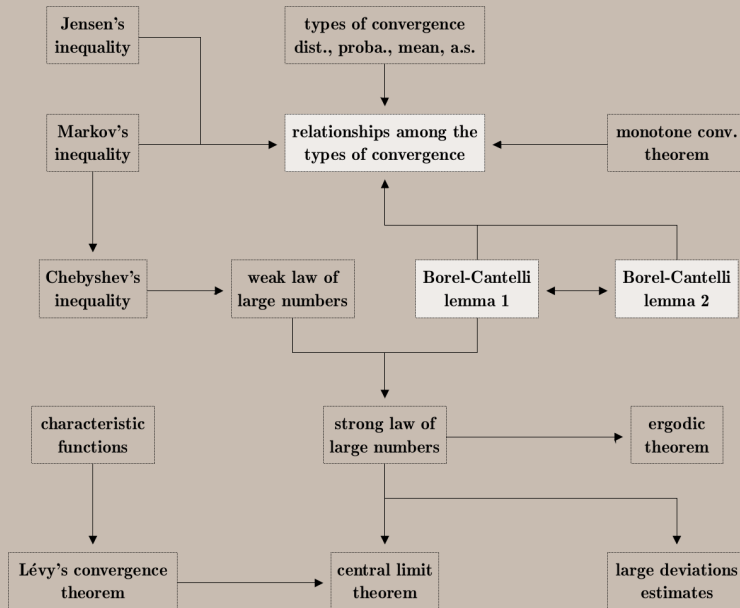
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Emile Borel



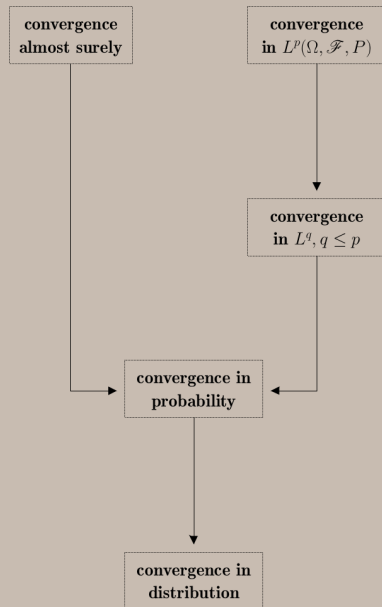
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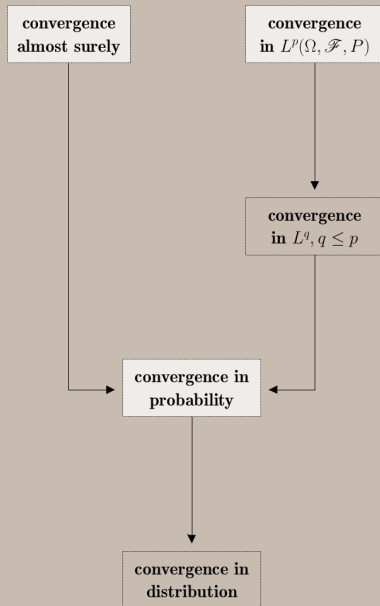


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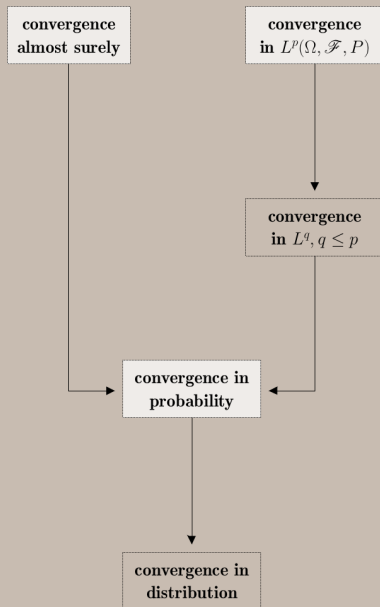


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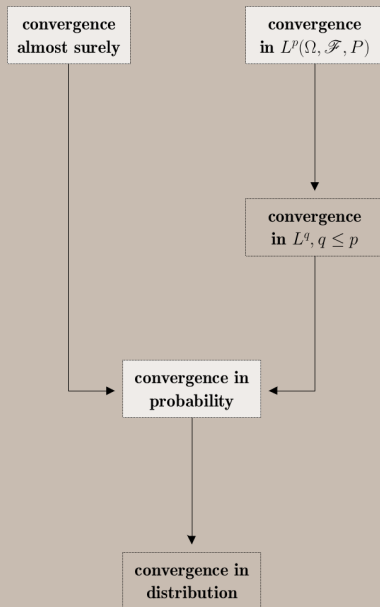
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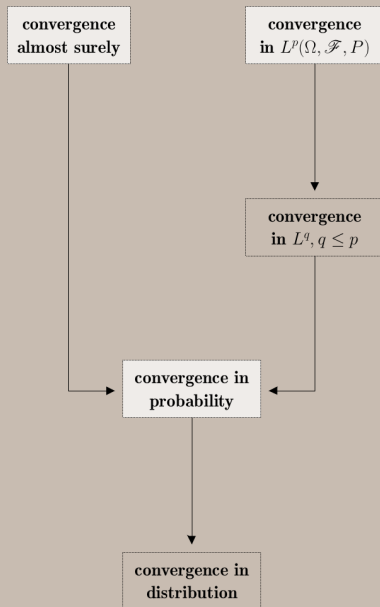
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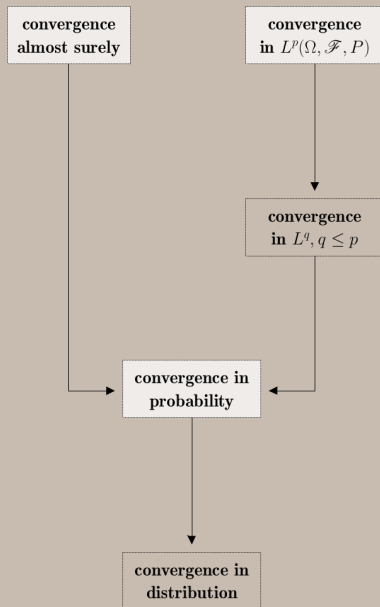
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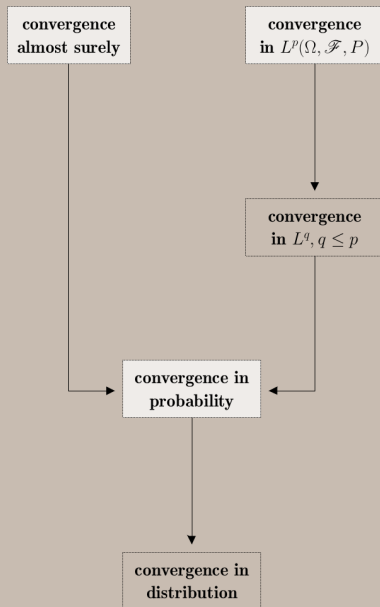
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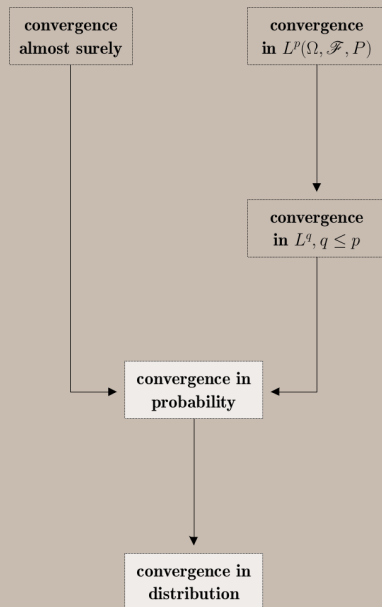
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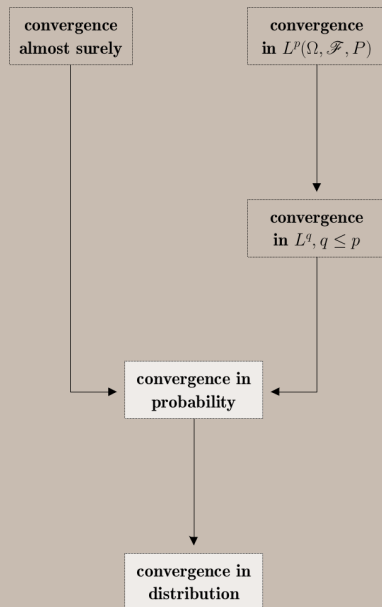
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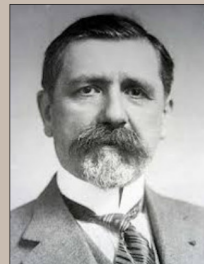
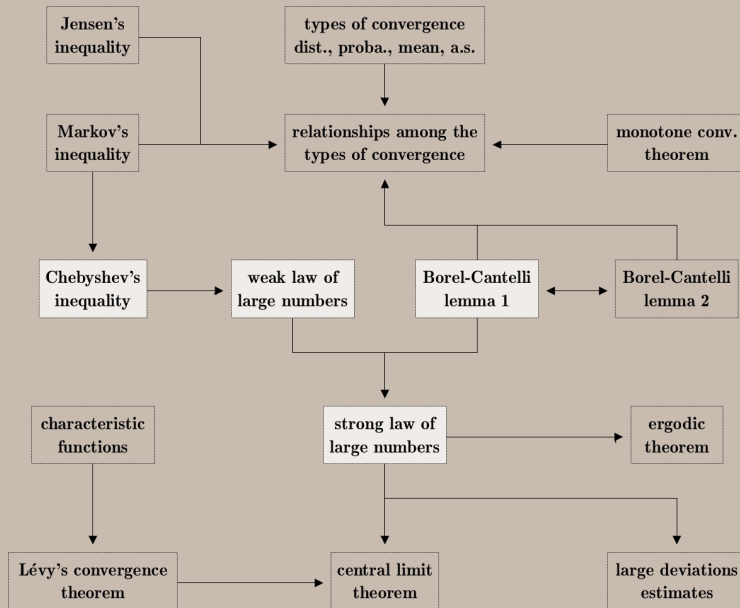
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Proof. Let X_n be independent Bernoulli(1/2) = coin flips.



Emile Borel



Francesco Cantelli

Law of large numbers. Let X_n be independent, identically distributed, with mean $\mu < \infty$.

weak: $S_n/n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu,$ **strong:** $S_n/n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \mu.$

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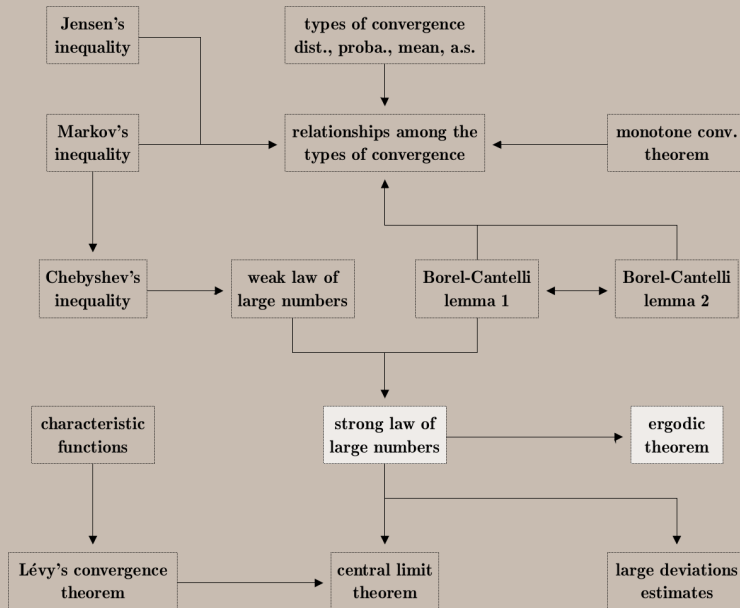
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$$\text{Conclusion: } |S_n/n - \mu| \leq |S_n/n - S_{k^2}/k^2| + |S_{k^2}/k^2 - \mu| \xrightarrow{a.s.} 0.$$



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John von Neumann

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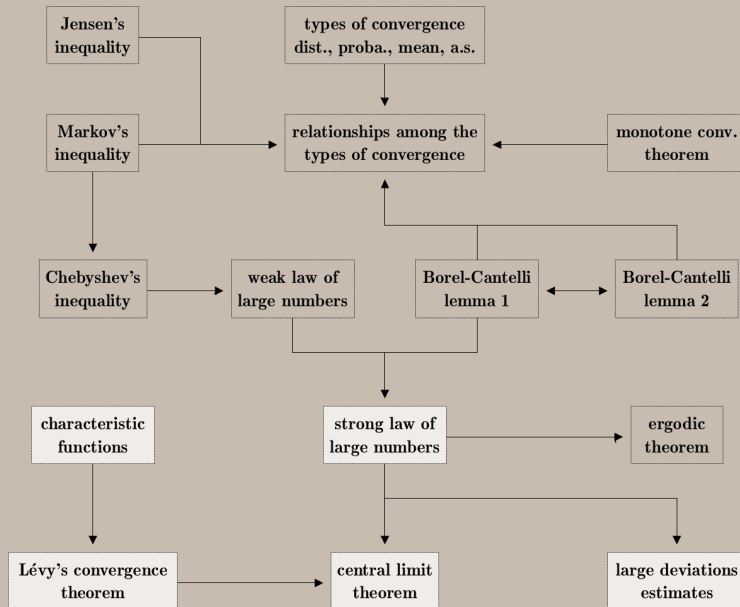
The ergodic theorem extends the strong law of large numbers.



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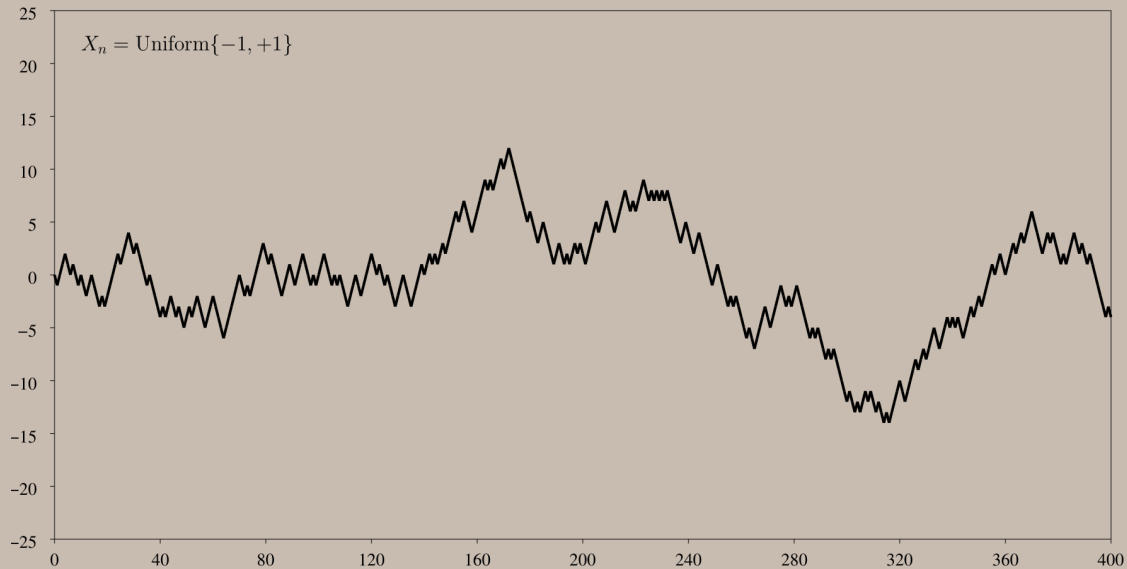
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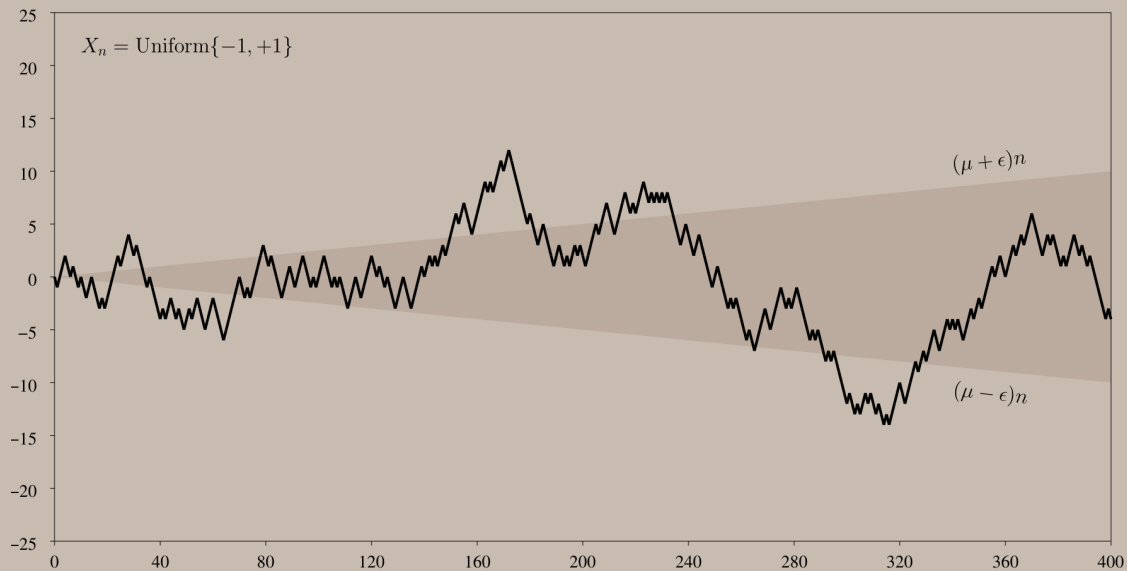


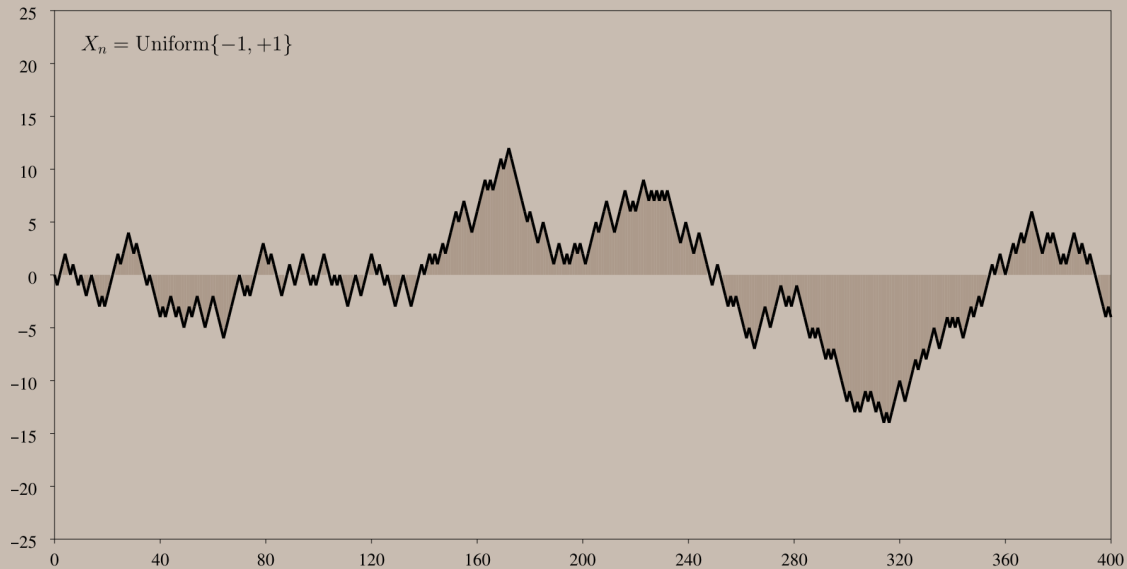
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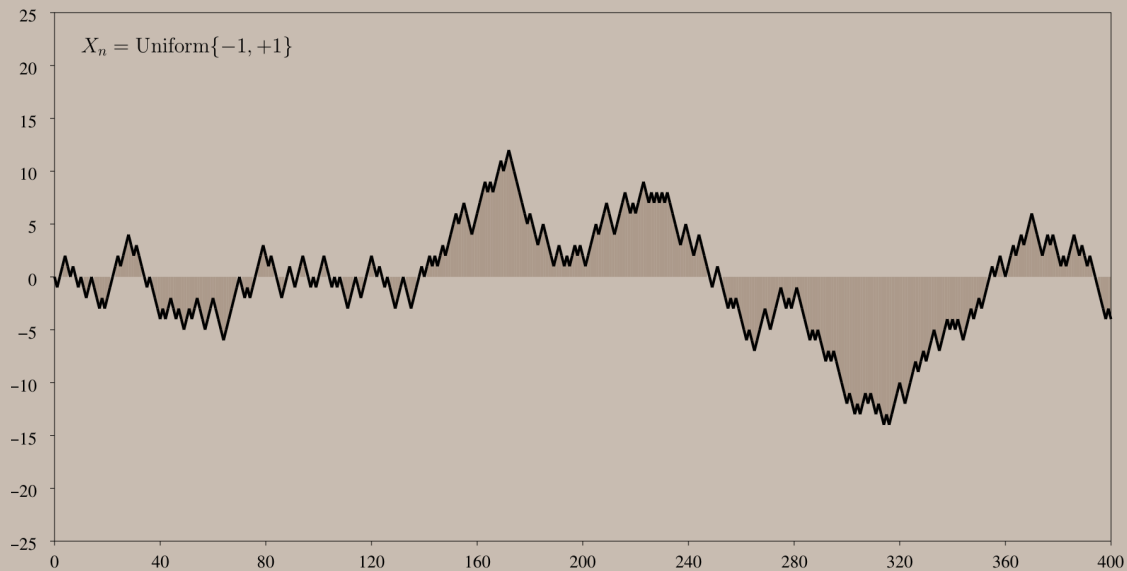
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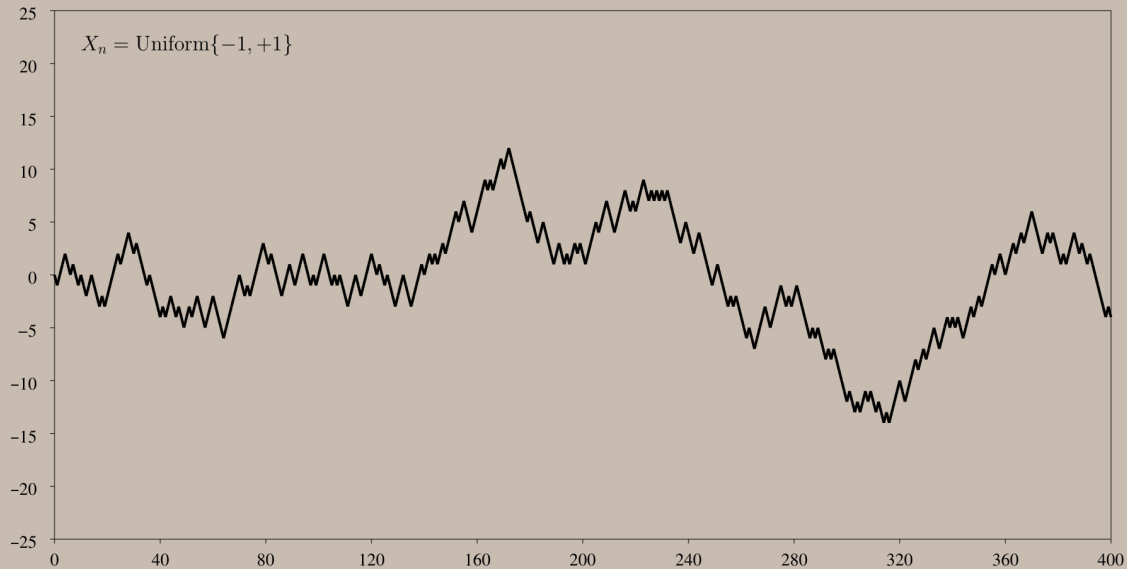
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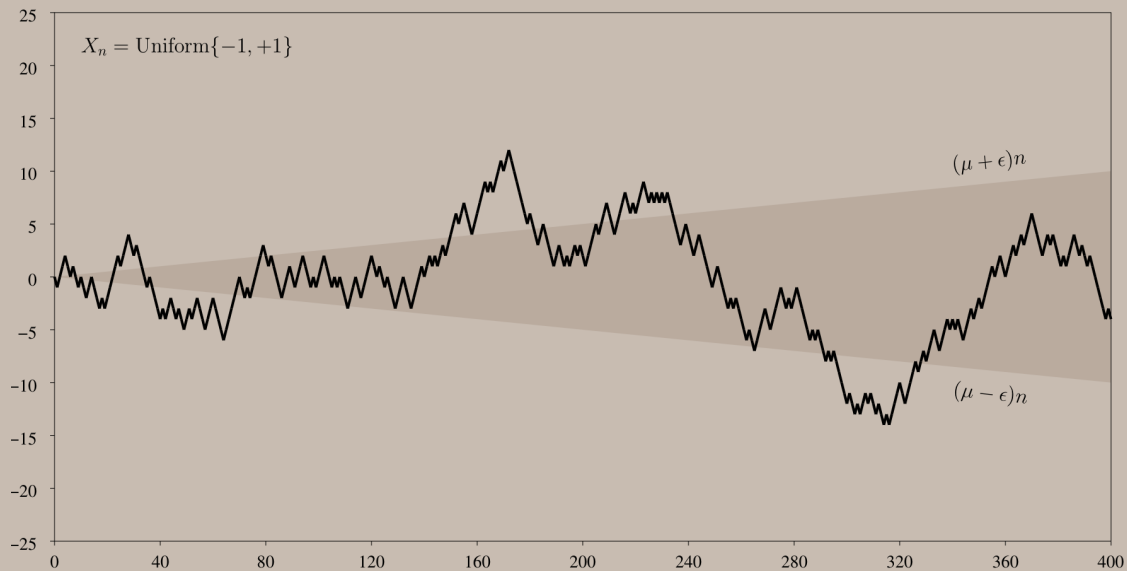


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Rate of convergence = ?

Large deviation estimates: Assume that $M(\theta) = E(e^{\theta X_i}) < \infty$ for all θ close to 0. Then, there is $\gamma = \gamma(\epsilon) > 0$ such that

$$P(|S_n/n - \mu| \geq \epsilon) \leq e^{-\gamma n}.$$

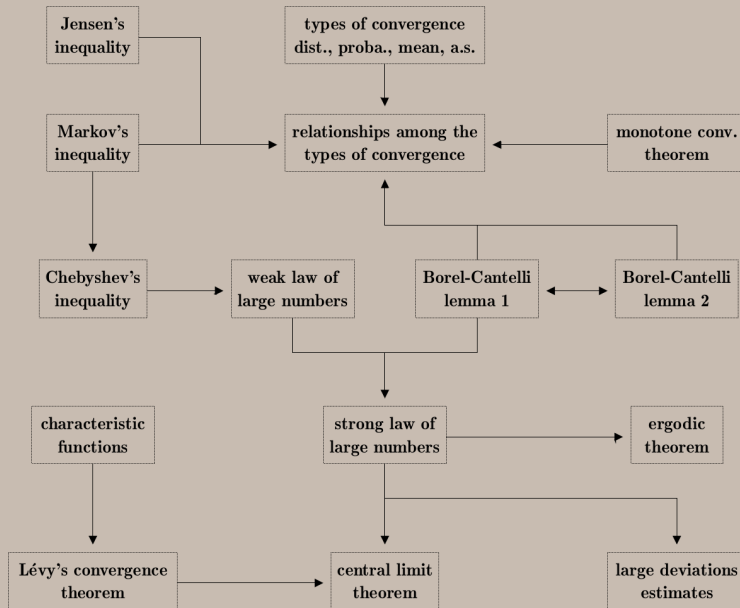


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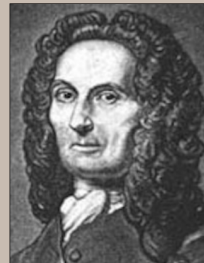


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Jakob Bernoulli



Abraham de Moivre